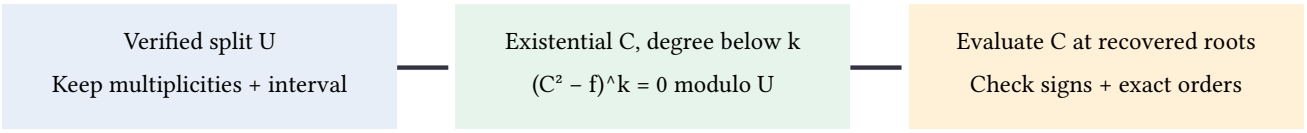
Uniform nilpotent square witness

2026-10-31 · Unreviewed derivation · No execution

For a verified split polynomial U of degree k, the ordinate rationality condition has an existential alternative with exponent k instead of an Euler exponent depending on p. This supplies a uniform candidate for every chart, including identically zero B.



Exact equivalence

Let p>3 be prime, $f=X^3+aX+b$ define a nonsingular curve, and U be monic, split over the prime field, of positive degree k. Every root α of U has square-or-zero f(α) if and only if there exists C of degree below k such that:

$$(C^2 - f)^k = 0 \text{ modulo } U.$$

Forward: choose a square root at each distinct root of U and interpolate C. Each linear factor divides C^2-f, so its kth power contains every multiplicity in U. Backward: evaluation at any root α gives $C(\alpha)^2=f(\alpha)$. This works for repeated roots and k≥p without division by multiplicities. U=1 is an explicit empty case.

Why exact square roots modulo U are too restrictive

Take $f=X^3+X$, $U=X^2$ over any prime p>3. This curve is nonsingular and the support is the rational branch point (0,0). Writing $C=c_0+c_1X$, the equation $C^2=f$ modulo X^2 forces $c_0=0$ and then 0=1 in the linear coefficient. It wrongly rejects. The nilpotent equation accepts C=0 because f^2 is divisible by X^2. This is a symbolic counterexample, not an executed trial.

Exact auxiliary multiplicity

If U has r distinct roots and q of their f-values are nonzero squares, there are exactly $\mathbf{2^q p^{(k-r)}}$ coefficient witnesses C. Evaluation from k coefficients onto r values is surjective with kernel dimension k−r. Each nonzero square offers two signs. A nonsquare anywhere gives zero witnesses. Repeated roots therefore create redundant coefficients that projected search must eliminate.

Arithmetic and recovery boundary

The quotient dimension and powering exponent are bounded by k, but k unknown field coefficients are added. Variable-U multiplication/reduction, bit ranges, carries and proof checking must still be counted. Field equations need prime-field membership: solving over an algebraic closure would change the predicate. No degree-of-regularity or runtime advantage follows from this equivalence.

After recovering verified roots of U, evaluate and square-check C to provide balanced root lifts. Retain the residual ratio for unbalanced signs and exact local multiplicities. A target outside U keeps its known ordinate and separate norm factor. Checking a supplied C needs no square-root search; finding C is part of solving.

Next comparison and provenance

Compare the complete Euler block, the uniform nilpotent-square block, and disjoint B-stratified guards, with identical interval, norm, chart and projected signed outputs. Retain every B=0 chart. Freeze fixtures before software execution; charge auxiliary projection and all reconstruction/checking.

The full proof is in proof.md. Chinese remainder background was read on 2026-10-31 from Stacks Project Lemma 10.15.4. Internal inputs are the unreviewed signed-recovery and conditional-square notes. This note asserts no novelty and changes no accepted finding or canonical graph.